IGCSE 0607 Practice Questions — E8.2

Vectors in two dimensions

Total Questions: 21 — **Total Marks:** 132 Marks

Table of Contents & Quick Navigation

- **E8.2** Vectors in two dimensions
  → **Paper 2 (Non-Calculator)** *(9 Qs)* · **Paper 4 (Calculator)** *(12 Qs)*

---

1 Vectors $\mathbf{p}$ and $\mathbf{q}$ are given as

$$\mathbf{p} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} \qquad \mathbf{q} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$$

(a) Work out

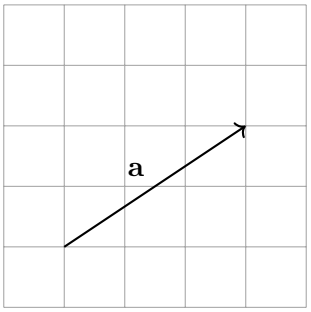
$$2\mathbf{p} + \mathbf{q}$$

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) Point A has coordinates $(1, 4)$.
The translation from A to B is given by the vector $\mathbf{p}$.
Work out the coordinates of B.

(..... , ...) **[1]**

2 The diagram shows a vector **a**.



(a) Write **a** as a column vector.

$\mathbf{a} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

(b) $\mathbf{b} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$.
Work out $2\mathbf{a} - \mathbf{b}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

3 The vectors **a** and **b** are given by

$\mathrm{a} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ and $\mathrm{b} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$ (0)

(a) Work out $\mathbf{a} + 2\mathbf{b}$ as a single column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) Point P has coordinates $(3, -1)$.
Point Q is translated from P by the vector **a**.
Find the coordinates of point Q.

(..... , ...) **[2]**

4 Given the vectors $\mathbf{p} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$.

(a) Work out $3\mathbf{p}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

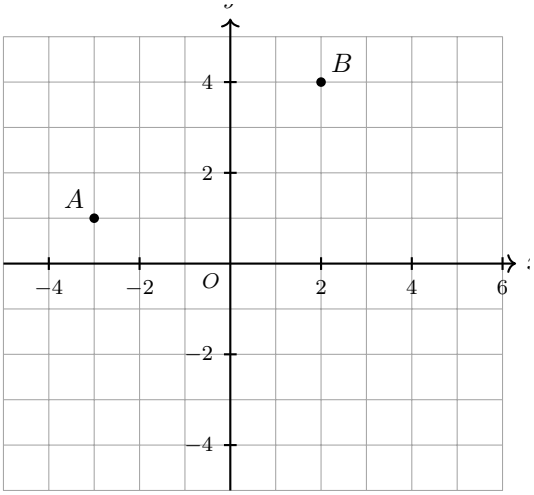
(b) Work out $\mathbf{p} - 2\mathbf{q}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(c) Point M is translated to point N by the vector $\mathbf{q}$.
Given that the coordinates of N are $(6, -1)$, find the coordinates of M.

(..... , ...) **[2]**

5 The diagram shows points $A(-3, 1)$ and $B(2, 4)$ on a coordinate grid.



(a) Write down $\vec{AB}$ as a column vector.

$\vec{AB} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) $\mathbf{p} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$.

Calculate

(i) $\mathbf{p} + 2\mathbf{q}$,

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

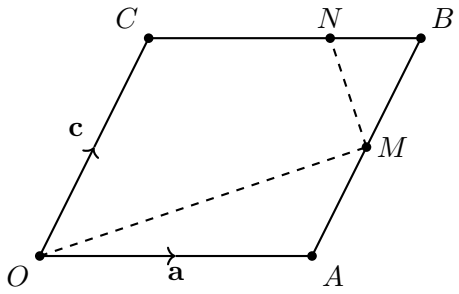
(ii) $3\mathbf{p} - \mathbf{q}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(c) Point E is such that $\vec{BE} = \begin{pmatrix} -4 \\ -6 \end{pmatrix}$.
Find the coordinates of E.

$E = (\ldots\ldots\ldots\ ,\ \ldots\ldots\ldots)$ **[2]**

6 The diagram shows a parallelogram $OABC$, where O is the origin.
$\vec{OA} = \mathbf{a}$ and $\vec{OC} = \mathbf{c}$.
M is the midpoint of AB and N is the point on BC such that $BN : NC = 2 : 1$.



(a) Find each of the following vectors in terms of **a** and **c**.
Give each answer in its simplest form.

(i) $\vec{OM}$

$\vec{OM} =$ ..... **[2]**

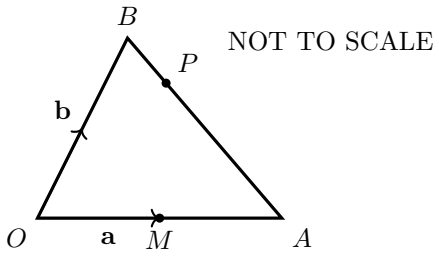
(ii) $\vec{MN}$

$\vec{MN} =$ ..... **[2]**

(b) $\vec{ON} = k\mathbf{a} + \mathbf{c}$.
Write down the value of k.

$k =$ ..... **[2]**

7 The diagram shows triangle OAB with $\vec{OA} = \mathbf{a}$ and $\vec{OB} = \mathbf{b}$.
M is the midpoint of OA, and P is the point on AB such that $AP : PB = 1 : 3$.



(a) Express each of the following vectors in terms of **a** and **b**, giving your answers in their simplest form:

(i) $\vec{AB}$,

$\vec{AB} =$ ..... **[1]**

(ii) $\vec{OP}$,

$\vec{OP} =$ ..... **[2]**

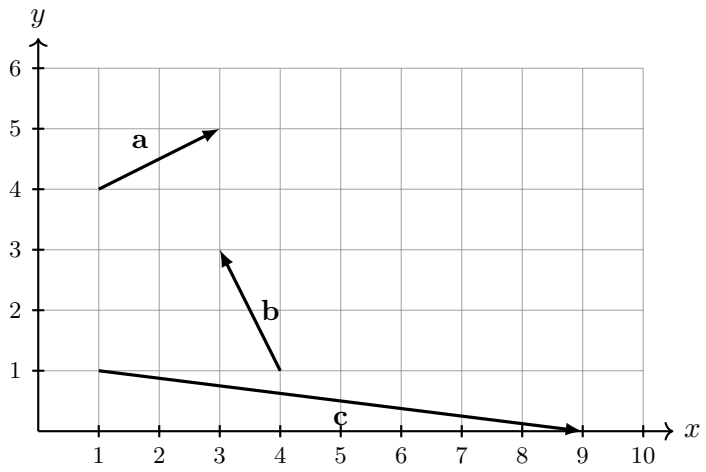
(iii) $\vec{MP}$.

$\vec{MP} =$ ..... **[2]**

(b) The line OP is extended to meet the line through B parallel to OA at the point C, where $\vec{BC} = k\mathbf{a}$ and $\vec{OC} = m\vec{OP}$.
Find the value of k and the value of m.

$k =$ ..... , $m =$ ..... **[4]**

8 The diagram shows vectors **a**, **b** and **c** plotted on a Cartesian coordinate grid.



(a) Write **c** as a column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

(b) Vector **c** can be written in the form $p\mathbf{a} + q\mathbf{b}$, where p and q are integers.
Find the value of p and the value of q.

$p =$ ..... , $q =$ ..... **[4]**

9 Vectors $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are given by

$$\mathbf{u} = \begin{pmatrix} 3 \\ m \end{pmatrix} \qquad \mathbf{v} = \begin{pmatrix} n \\ -2 \end{pmatrix} \qquad \mathbf{w} = \begin{pmatrix} 10 \\ 7 \end{pmatrix}$$

(a) Given that $2\mathbf{u} - \mathbf{v} = \mathbf{w}$, find the value of m and the value of n.

$m =$ ..... , $n =$ ..... **[3]**

(b) Vector $\mathbf{z} = \begin{pmatrix} k \\ 12 \end{pmatrix}$.
Given that $\mathbf{z}$ is parallel to $\mathbf{u}$, find an expression for k in terms of m.

$k =$ ..... **[3]**

10 $\mathbf{p} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ $\mathbf{q} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$

(a) Work out $3\mathbf{p} - 2\mathbf{q}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) Point A has coordinates $(3, -1)$.
$\vec{AB} = \mathbf{p} + \mathbf{q}$.
Find the coordinates of B.

(..... , ...) **[2]**

11 Vectors $\mathbf{p}$ and $\mathbf{q}$ are given as:

$$\mathbf{p} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} \qquad \mathbf{q} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$$

(a) Work out $3\mathbf{p}$.

$$3\mathbf{p} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$$ **[1]**

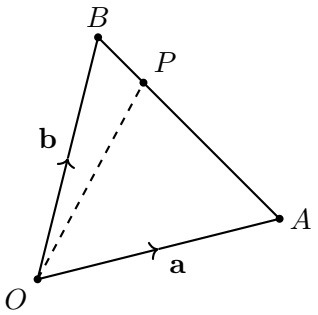
(b) Work out $\mathbf{p} - 2\mathbf{q}$.

$$\mathbf{p} - 2\mathbf{q} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$$ **[2]**

(c) Point A has coordinates $(3, -1)$.
Point B is the image of A after a translation by the vector $\mathbf{p}$.
Work out the coordinates of B.

(..... , ...) **[2]**

12 In triangle OAB, $\vec{OA} = \mathbf{a}$ and $\vec{OB} = \mathbf{b}$.
P is the point on AB such that $AP : PB = 3 : 1$.



(a) Find $\vec{OP}$ in terms of **a** and **b**.
Give your answer in its simplest form.

$\vec{OP} =$ ..... **[3]**

(b) M is the midpoint of OA.
Find $\vec{MP}$ in terms of **a** and **b**.
Give your answer in its simplest form.

$\vec{MP} =$ ..... **[2]**

13 Vectors **u**, **v** and **w** are given by:

$$\mathbf{u} = \begin{pmatrix} 3 \\ -2 \end{pmatrix} \qquad \mathbf{v} = \begin{pmatrix} x \\ 5 \end{pmatrix} \qquad \mathbf{w} = \begin{pmatrix} -1 \\ y \end{pmatrix}$$

(a) Given that $2\mathbf{u} + \mathbf{v} = \mathbf{w}$, find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[3]**

(b) Point M has coordinates $(4, 7)$ and $\vec{MN} = \begin{pmatrix} -2 \\ 6 \end{pmatrix}$.

(i) Find the coordinates of N.

(..... , ...) **[2]**

(ii) Write down the vector $\vec{NM}$.

$\vec{NM} =$ ( ) **[1]**

14 **(a)** $\mathbf{u} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} -1 \\ y \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} 10 \\ 6 \end{pmatrix}$.

Given that $3\mathbf{u} + k\mathbf{v} = \mathbf{w}$, work out the value of k and the value of y.

$k =$ ..... , $y =$ ..... **[3]**

(b) Points P, Q and R have coordinates $(2, -1)$, $(5, 5)$ and $(11, 17)$ respectively.

(i) Write $\vec{PQ}$ as a column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

(ii) Show that P, Q and R lie on a straight line.

[2]

15 The coordinates of point A are $(2, 3)$ and the coordinates of point B are $(-1, 5)$.

(a) Write down the column vector $\vec{AB}$.

$\vec{AB} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

(b) Point C is such that $\vec{BC} = 3\vec{AB}$.
Find the coordinates of C.

(..... , ...) **[3]**

(c) $\mathbf{p} = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$.
Find the column vector $\mathbf{q}$ such that $\vec{AB} + \mathbf{q} = \mathbf{p}$.

$\mathbf{q} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

16 Points A, B and C have coordinates $(1, -2)$, $(7, 10)$ and $(4, 4)$ respectively.

(a) Write $\vec{AB}$ as a column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) Show that $\vec{AC} = \frac{1}{2}\vec{AB}$.

[2]

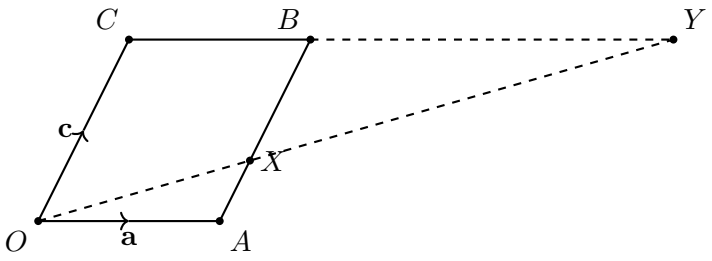
(c) Point C is translated by the vector $\begin{pmatrix} -5 \\ 2 \end{pmatrix}$ to point D.
Find the coordinates of D.

(..... , ...) **[2]**

(d) Work out the vector $\vec{BD}$ as a column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

17 $OABC$ is a parallelogram.
$\vec{OA} = \mathbf{a}$ and $\vec{OC} = \mathbf{c}$.
X is a point on AB such that $AX : XB = 1 : 2$.
Y is a point on the line CB extended such that $CB : BY = 1 : 2$.



(a) Find $\vec{OX}$ in terms of **a** and **c**.

$\vec{OX} =$ ..... **[2]**

(b) Find $\vec{OY}$ in terms of **a** and **c**.

$\vec{OY} =$ ..... **[2]**

(c) Show that the points O, X and Y lie on a straight line.

[2]

(d) Find the ratio $OX : XY$.

$OX : XY =$ ..... **[1]**

18 The position vectors of points A, B, and C relative to an origin O are given by

$= \begin{pmatrix} 2 \\ -1 \end{pmatrix}$, $= \begin{pmatrix} 6 \\ 7 \end{pmatrix}$, and $= \begin{pmatrix} k+1 \\ 3k-2 \end{pmatrix}$

where k is a constant.

(a) Find $\vec{AB}$ as a column vector.

$\vec{AB} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) Given that points A, B, and C are collinear, find the value of k.

$k =$ ..... **[3]**

(c) Using your value of k from part (b), write down $\vec{AC}$ in terms of $\vec{AB}$.

$\vec{AC} =$ ..... **[2]**

19 Given the vectors $\mathbf{u} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$.

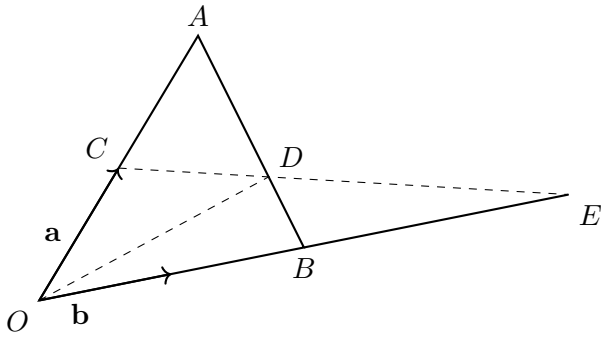
(a) Point P has coordinates $(5, -1)$.
The vector $\vec{PQ} = 2\mathbf{u} - \mathbf{v}$.
Find the coordinates of Q.

(..... , ...) **[3]**

(b) Point R has coordinates $(14, 3)$.
The vector $\vec{PR}$ can be written in the form $h\mathbf{u} + k\mathbf{v}$.
Work out the values of the constants h and k.

$h =$ ..... , $k =$ ..... **[5]**

20 The diagram shows a triangle OAB.
O is the origin, $\vec{OA} = \mathbf{a}$ and $\vec{OB} = \mathbf{b}$.
C is the midpoint of OA.
D is a point on AB such that $AD : DB = 2 : 1$.
E is a point on OB extended such that $\vec{OE} = 2\mathbf{b}$.



NOT TO SCALE

(a) Find $\vec{AB}$ in terms of **a** and **b**.

$\vec{AB} =$ ..... **[1]**

(b) Find $\vec{OD}$ in terms of **a** and **b**.
Give your answer in its simplest form.

$\vec{OD} =$ ..... **[3]**

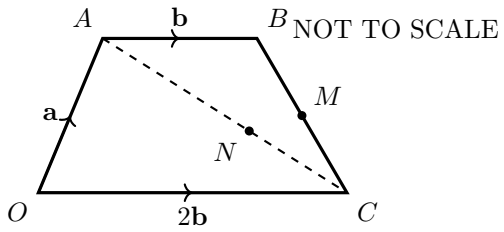
(c) Find $\vec{CE}$ in terms of **a** and **b**.

$\vec{CE} =$ ..... **[2]**

(d) Show that the points C, D and E lie on a straight line.

[2]

21 The diagram shows trapezium $OABC$ with $\vec{OA} = \mathbf{a}$, $\vec{AB} = \mathbf{b}$, and $\vec{OC} = 2\mathbf{b}$.
M is the midpoint of BC, and N is the point on AC such that $AN : NC = 3 : 2$.



(a) Express each of the following vectors in terms of **a** and **b**, giving your answers in their simplest form:

(i) $\vec{OB}$,

$\vec{OB} =$ ..... **[1]**

(ii) $\vec{BC}$,

$\vec{BC} =$ ..... **[2]**

(iii) $\vec{OM}$.

$\vec{OM} =$ ..... **[2]**

(b) Express $\vec{ON}$ in terms of **a** and **b**, giving your answer in its simplest form.

$\vec{ON} =$ ..... **[3]**

(c) Use your answers to part (a)(iii) and part (b) to

(i) explain why O, N, and M lie on a straight line,

[2]

(ii) write down the ratio $ON : NM$ in its simplest form.

$ON : NM =$ ..... **[2]**